(iii) $\operatorname{cosec} 39^\circ = \operatorname{cosec}(90^\circ - 51^\circ)$ (since $39^\circ = 90^\circ - 51^\circ$)

RHS is of the form $\operatorname{cosec}(90^\circ - \theta) = \sec\theta$

Therefore $\operatorname{cosec} 39^\circ = \sec 51^\circ$

Example 6.8

Evaluate: (i) $\dfrac{\sin 49^\circ}{\cos 41^\circ}$ (ii) $\dfrac{\sec 63^\circ}{\operatorname{cosec} 27^\circ}$

Solution

(i) $\dfrac{\sin 49^\circ}{\cos 41^\circ}$

$\sin 49^\circ = \sin(90^\circ - 41^\circ) = \cos 41^\circ$, since $49^\circ + 41^\circ = 90^\circ$ (complementary),

Hence on substituting $\sin 49^\circ = \cos 41^\circ$ we get, $\dfrac{\cos 41^\circ}{\cos 41^\circ} = 1$

(ii) $\dfrac{\sec 63^\circ}{\operatorname{cosec} 27^\circ}$

$\sec 63^\circ = \sec(90^\circ - 27^\circ) = \operatorname{cosec} 27^\circ$, here 63° and 27° are complementary angles.

we have $\dfrac{\sec 63^\circ}{\operatorname{cosec} 27^\circ} = \dfrac{\operatorname{cosec} 27^\circ}{\operatorname{cosec} 27^\circ} = 1$

Example 6.9

Find the values of (i) $\tan 7^\circ \tan 23^\circ \tan 60^\circ \tan 67^\circ \tan 83^\circ$

(ii) $\dfrac{\cos 35^\circ}{\sin 55^\circ} + \dfrac{\sin 12^\circ}{\cos 78^\circ} - \dfrac{\cos 18^\circ}{\sin 72^\circ}$

Solution

(i) $\tan 7^\circ \tan 23^\circ \tan 60^\circ \tan 67^\circ \tan 83^\circ$

$= \tan 7^\circ \tan 83^\circ \tan 23^\circ \tan 67^\circ \tan 60^\circ$ (Grouping complementary angles)

$= \tan 7^\circ \tan(90^\circ - 7^\circ) \tan 23^\circ \tan(90^\circ - 23^\circ) \tan 60^\circ$

$= (\tan 7^\circ . \cot 7^\circ)(\tan 23^\circ . \cot 23^\circ) \tan 60^\circ$

$= (1) \times (1) \times \tan 60^\circ$

$= \tan 60^\circ \quad = \sqrt{3}$

(ii) $\dfrac{\cos 35^\circ}{\sin 55^\circ} + \dfrac{\sin 12^\circ}{\cos 78^\circ} - \dfrac{\cos 18^\circ}{\sin 72^\circ}$

$= \dfrac{\cos(90^\circ - 55^\circ)}{\sin 55^\circ} + \dfrac{\sin(90^\circ - 78^\circ)}{\cos 78^\circ} - \dfrac{\cos(90^\circ - 72^\circ)}{\sin 72^\circ}$ $\left[\begin{array}{l}\text{since} \\ \cos 35^\circ = \cos(90^\circ - 55^\circ) \\ \sin 12^\circ = \sin(90^\circ - 78^\circ) \\ \cos 18^\circ = \cos(90^\circ - 72^\circ)\end{array}\right]$

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